Test 1 [Make up]

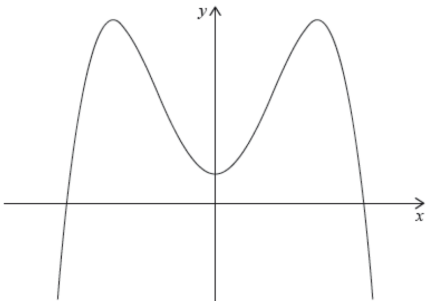
EXERCISE 1

1. Given $y = x^2 + 5x - 2$ find $\frac{dy}{dx}$ from first principles.
2. Hence find the point on the graph at which the tangent has gradient -3.

EXERCISE 2 Differentiate $y = \sqrt{1 - x^2}$ with respect to x.

EXERCISE 3 The gradient of the tangent line to the graph of $f(x) = 2x^2 + kx - 3$ at $x = -1$ is 1. Find the value of k.

EXERCISE 4



Consider the function $f(x) = -x^4 + ax^2 + 5$ where a is a constant. Part of the graph of $y = f(x)$ is shown.

It is known that at the point where $x = 2$, the tangent to the graph of $y = f(x)$ is horizontal. There are two other points on the graph of $y = f(x)$ at which the tangent is horizontal.

1. Write down the y-intercept of the graph.
2. Find $f'(x)$.
3. **a.** Show that $a = 8$.

   b. Find $f(2)$.

   ONLY HL from now on
4. **a.** Write down the x-coordinates of these two points.

   b. Write down the intervals where the gradient of the graph of $y = f(x)$ is positive.
5. Write down the range of $f(x)$.
6. Write down the number of possible solutions to the equation $f(x) = 5$.